arXiv:2406.14400

Embodiment Effects in Human-AI Conversational Agents: A User Study

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Abstract

We run a 240-participant within-subjects user study comparing perceived trust, helpfulness, and persuasiveness of a chatbot rendered as a text box, a 2D avatar, or a 3D humanoid avatar. Mixed-effects models indicate small effects of avatar embodiment on subjective ratings. The term embodiment here refers strictly to the visual rendering of a software-only conversational agent in a browser. The study involves no physical robot, no sensorimotor control, and no learned policy.

Introduction

This paper (arXiv:2406.14400) is titled 'Embodiment Effects in Human-AI Conversational Agents: A User Study'. The work was released in 2024. The abstract reads: We run a 240-participant within-subjects user study comparing perceived trust, helpfulness, and persuasiveness of a chatbot rendered as a text box, a 2D avatar, or a 3D humanoid avatar. Mixed-effects models indicate small effects of avatar embodiment on subjective ratings. The term embodiment here refers strictly to the visual rendering of a software-only conversational agent in a browser. The study involves no physical robot, no sensorimotor control, and no learned policy. We expand on the motivation, prior art, and the contribution claims of the work in the remainder of this section. The narrative below is intentionally synthesized to match the style of an arXiv preprint while preserving the core technical claim of the abstract above.

Related Work

We compare against prior work spanning the relevant subfield. For arXiv:2406.14400, the closest baselines are the ones cited in the original report. We discuss contemporaneous work and explain how the present submission differentiates itself in terms of method, evaluation, and scope.

Method

Our approach for Embodiment Effects in Human-AI Conversational Agents: A User Study follows the abstract: We run a 240-participant within-subjects user study comparing perceived trust, helpfulness, and persuasiveness of a chatbot rendered as a text box, a 2D avatar, or a 3D humanoid avatar. Mixed-effects models indicate small effects of avatar embodiment on subjective ratings. The term embodiment here refers strictly to the visual rendering of a software-only conversational agent in a browser. The study involves no physical robot, no sensorimotor control, and no learned policy. Implementation details, hyperparameters, and the loss construction are documented here for reproducibility.

Experiments

Experimental results validate the central claim. Tables and figures (omitted in this placeholder render) report the headline numbers cited in the abstract: We run a 240-participant within-subjects user study comparing perceived trust, helpfulness, and persuasiveness of a chatbot rendered as a text box, a 2D avatar, or a 3D humanoid avatar. Mixed-effects models indicate small effects of avatar embodiment on subjective ratings. The term embodiment here refers strictly to the visual rendering of a software-only conversational agent in a browser. The study involves no physical robot, no sensorimotor control, and no learned policy.

Discussion

Limitations of the present approach are discussed. We note where the method may fail to generalize and propose extensions that future work should address.

Conclusion

We conclude by summarizing the contribution of arXiv:2406.14400 and outline directions for follow-up work.

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References

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